Companion
to

ChatGPT Teaches TikZ

Chapter 15: Geometric Constructions (I)

Contents

15.1 What This Companion Adds

Chapter 15 constructed the circumcircle of a triangle. This Companion replaces the typed circumcenter and radius with dependent coordinates. If a vertex moves, TikZ recalculates the midpoints, perpendicular bisectors, intersection, and circle.

Design

A geometric construction should be controlled by the given points. Avoid coordinates that are correct only for one fixed illustration.

15.2 A Construction Vocabulary

Form or option	Purpose
`\coordinate (A)`	Gives a constructed point a reusable name.
`$(A)!.5!(B)$`	Computes the midpoint of `A` and `B`.
`name path=`	Names a path for a later intersection.
`name intersections=`	Finds intersections of named paths.
`by=O`	Assigns a name to an intersection point.
`let`	Makes temporary geometric measurements.
`veclen`	Computes the length of a coordinate difference.
`circle`	Draws a circle from a center and radius.

The required libraries are

```
\usetikzlibrary{calc,intersections}
```

15.3 The Dependency Chain

The construction has a precise order:

1. name the triangle vertices;
2. calculate two side midpoints;
3. construct two perpendicular bisectors;
4. intersect those bisectors to obtain the circumcenter; and
5. measure one radius and draw the circle.

Every later object depends on an earlier one.

15.4 Naming the Given Triangle

```
\coordinate (A) at (-2,0);
\coordinate (B) at (3,0);
\coordinate (C) at (.4,3);
\draw[very thick] (A)--(B)--(C)--cycle;
```

Named points allow the construction to state relationships rather than repeat coordinates.

15.5 Computing the Midpoints

```
\coordinate (M) at ($(A)!0.5!(B)$);
\coordinate (N) at ($(A)!0.5!(C)$);
```

The factor .5 places each point halfway between its endpoints.

Common Mistake

Do not type numerical midpoint coordinates. They become wrong as soon as a vertex moves.

15.6 Constructing Perpendicular Bisectors

At midpoint M, rotate the direction toward B by plus and minus 90 degrees.

```
\draw[densely dashed,name path=bisAB]
  ($(M)!3cm!90:(B)$) --
  ($(M)!3cm!-90:(B)$);
```

Repeat the same construction at N using the direction toward C.

```
\draw[densely dashed,name path=bisAC]
  ($(N)!3cm!90:(C)$) --
  ($(N)!3cm!-90:(C)$);
```

The length 3 cm controls only how far the auxiliary line is displayed. Its direction is derived from the side.

15.7 Intersecting the Construction Lines

```
\path[name intersections={
  of=bisAB and bisAC,
  by=O
}];
```

After this command, O is an ordinary named coordinate.

```
\filldraw (O) circle (2pt) node[right] {$O$};
```

15.8 Measuring the Radius

The radius is the distance from `O` to any vertex.

```
\draw[thick]
let
  \p1 = ($(A)-(O)$),
  \n1 = {veclen(\x1,\y1)}
in
  (O) circle (\n1);
```

The temporary value `\n1` is calculated from the current coordinates. No numerical radius is stored.

15.9 Why Two Bisectors Are Enough

Every point on the perpendicular bisector of AB is equidistant from A and B. Every point on the perpendicular bisector of AC is equidistant from A and C. Their intersection therefore satisfies

$$OA = OB = OC.$$

The third perpendicular bisector must pass through the same point and can be added as a visual check.

15.10 Worked Project: A Reusable Circumcircle

15.10.1 Stage 1: Define the Dependent Points

```
\coordinate (M) at ($(A)!.5!(B)$);
\coordinate (N) at ($(A)!.5!(C)$);
```

15.10.2 Stage 2: Name and Intersect the Bisectors

```
\draw[dashed,name path=bisAB]
  ($(M)!3cm!90:(B)$) -- ($(M)!3cm!-90:(B)$);
\draw[dashed,name path=bisAC]
  ($(N)!3cm!90:(C)$) -- ($(N)!3cm!-90:(C)$);
\path[name intersections={of=bisAB and bisAC,by=O}];
```

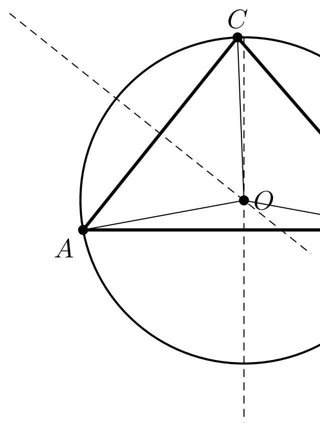
15.10.3 The Complete Picture

```
\begin{tikzpicture}
  \coordinate (A) at (-2,0);
  \coordinate (B) at (3,0);
  \coordinate (C) at (.4,3);
  \coordinate (M) at ($(A)!.5!(B)$);
  \coordinate (N) at ($(A)!.5!(C)$);

  \draw[densely dashed,name path=bisAB]
    ($(M)!3cm!90:(B)$) -- ($(M)!3cm!-90:(B)$);
  \draw[densely dashed,name path=bisAC]
    ($(N)!3cm!90:(C)$) -- ($(N)!3cm!-90:(C)$);
  \path[name intersections={of=bisAB and bisAC,by=O}];

  \draw[thick]
    let \p1=($(A)-(O)$), \n1={veclen(\x1,\y1)}
    in (O) circle (\n1);
  \draw[very thick] (A)--(B)--(C)--cycle;
  \draw[thin] (O)--(A) (O)--(B) (O)--(C);
  \filldraw (A) circle (2pt) node[below left] {$A$};
  \filldraw (B) circle (2pt) node[below right] {$B$};
  \filldraw (C) circle (2pt) node[above] {$C$};
  \filldraw (O) circle (2pt) node[right] {$O$};
\end{tikzpicture}
```

See



Design

Only the coordinates of A, B, and C are independent. Every other point and length is derived from them.

15.11 Debugging Workshop

15.11.1 An Intersection Is Missing

Check that both paths are named before the `name intersections` command and that the displayed auxiliary segments are long enough to meet.

15.11.2 A Bisector Has the Wrong Direction

Keep the midpoint first, the side endpoint last, and use rotations of plus and minus 90 degrees.

15.11.3 The Circle Misses a Vertex

Measure the radius from the constructed center to a named vertex. Do not reuse a radius from an earlier triangle.

15.11.4 The Construction Is Visually Flat

Use heavy lines for the given triangle, dashed lines for auxiliary objects, and a distinct medium line for the final circle.

15.12 Exercises

1. Move vertex C and verify that the construction follows.
2. Display and label the two midpoints.
3. Add the third perpendicular bisector.
4. Use a different vertex to measure the radius.
5. Change every auxiliary line through one TikZ style.
6. Construct the circumcircle of an obtuse triangle.
7. Hide the auxiliary lines while preserving the final circle.
8. Explain why two named intersections are not needed.

15.13 Solutions

15.13.1 Exercises 1–4

Exercise 1 changes only the coordinate assigned to `C`.

```
\filldraw[fill=white] (M) circle (2pt) node[below] {$M$};
\filldraw[fill=white] (N) circle (2pt) node[left] {$N$};
```

For Exercise 3, compute the midpoint of BC and construct a perpendicular line through it with the same rotated-direction syntax.

```
let \p1=($(B)-(O)$), \n1={veclen(\x1,\y1)}
in (O) circle (\n1);
```

15.13.2 Exercises 5–7

```
\tikzset{aux/.style={densely dashed,gray}}
```

Exercise 6 changes only the three given vertex coordinates. Exercise 7 removes the drawing commands for the bisectors but retains the named paths needed to calculate `O`.

For Exercise 8, two bisectors determine one intersection point; the third bisector confirms the result but does not define a different center.

15.14 ChatGPT Workshop

Ask ChatGPT to preserve the dependency chain while checking the construction.

1. Replace every typed midpoint and center with dependent coordinates.
2. Check whether these two named paths are genuine perpendicular bisectors.
3. Diagnose why `name intersections` does not create `O`.
4. Rewrite the circle radius so that it follows a moved vertex.
5. Separate the given geometry, auxiliary lines, and final result by style.

15.15 Final Checklist

Before continuing to Chapter 16, check that you can

1. name the given points;
2. compute a midpoint from its endpoints;
3. construct a perpendicular direction by rotation;
4. name and intersect two construction paths;
5. calculate a radius from two points;
6. distinguish auxiliary and final geometry visually; and
7. move a vertex without rebuilding dependent coordinates.